

Assignment:-

1) Error reading of first meter

$$= \text{FSR} \times \text{accuracy} = 20 \times \frac{\pm 0.1}{100} = \pm 0.02$$

Error in reading of second meter

$$= 10 \times \frac{\pm 0.2}{100} = \pm 0.02$$

Error in reading of third meter

$$= 5 \times \frac{\pm 0.5}{100} = \pm 0.025$$

Error in reading of fourth meter

$$= 1 \times \frac{\pm 1.00}{100} = \pm 0.01$$

Fourth meter has least error.

2) Thermocouple, instrument read rms value

$$I_{\text{rms}} = \sqrt{2^2 + \left(\frac{\sqrt{2}}{\sqrt{2}}\right)^2 + \left(\frac{2\sqrt{2}}{\sqrt{2}}\right)^2} \text{ A}$$

$$= \sqrt{4+4} = 3 \text{ A}$$

For class 1 meter, accuracy is 1% for 5A range.

For 3A, accuracy will be $\frac{5}{3}\% = 1.67\%$.

$$3) R_1 = 10 \pm 10 \times \frac{5}{100} = 9.5 \Omega \text{ to } 10.5 \Omega$$

$$R_2 = 5 \pm 5 \times \frac{10}{100} = 4.5 \Omega \text{ to } 5.5 \Omega$$

$$R_p = \frac{R_1 R_2}{R_1 + R_2}$$

$$R_p = \frac{9.5 \times 4.5}{9.5 + 4.5} \text{ to } \frac{10.5 \times 5.5}{10.5 + 5.5}$$

option A
is (1)

4.) Full scale reading of all three = 100
 $n = 80$; $y = 20$; $z = 50$

$$\delta n = \pm 0.5\% \text{ of reading} = \pm \frac{0.5 \times 80}{100} = \pm 0.4$$

$$\delta y = \pm 1\% \text{ of full reading} = \pm \frac{1 \times 100}{100} = \pm 1$$

$$\delta z = \pm 1.5\% \text{ of reading} = \pm \frac{1.5 \times 50}{100} = \pm 0.75$$

$$w = \frac{ny}{z}$$

$$\frac{\delta w}{w} = \frac{\delta n}{n} + \frac{\delta y}{y} - \frac{\delta z}{z}$$

$$\frac{\delta w}{w} = \pm \left(\frac{0.4}{80} + \frac{1}{20} + \frac{0.75}{100} \right) \times 100 = 1\%$$

5.) $\frac{dG_1}{G_1} = \epsilon_1$, $\frac{dG_2}{G_2} = \epsilon_2$ and $\frac{dG_3}{G_3} = \epsilon_3$

$$y = \frac{G_1 G_2}{G_3} \times$$

$n = \text{input}$

$$\ln y = \ln G_1 + \ln G_2 - \ln G_3 + \ln x$$

$$\frac{dy}{y} = \frac{dG_1}{G_1} + \frac{dG_2}{G_2} - \frac{dG_3}{G_3} + \frac{dn}{n}$$

$$\frac{dn}{n} = 0$$

$$\frac{dy}{y} = \epsilon_1 + \epsilon_2 - \epsilon_3$$

6.) $R_1 = 50 \Omega$, $R_2 = 65 \Omega$, $R_3 = 100 \pm 5\% \Omega$

$$R_3 = 100 \pm 5\% \Omega$$

$$= 100 + 100 \times \frac{5}{100} = 105 \Omega$$

$$= 100 - 100 \times \frac{5}{100} = 95 \Omega$$

$$\frac{R_1}{R_3} = \frac{R_2}{R_n}$$

$$R_n = \frac{R_2 R_3}{R_1}$$

$$(i) R_3 = 105 \Omega$$

$$R_n = \frac{105 \times 65}{50} = 136.50 \Omega$$

$$(ii) R_3 = 95 \Omega$$

$$R_n = \frac{95 \times 65}{50} = 123.50 \Omega$$

$$7) V = 220 \pm 1 \%$$

$$I = 5 \pm 1 \%$$

$$W = 555 \pm 2 \%$$

$$P.f = \cos(\phi) = \frac{W}{V \cdot I}$$

$$P.f = \cos \phi$$

$$= 555 \pm 2 \%$$

$$\frac{(220 \pm 1\%)(5 \pm 1\%)}{220 \times 5} = \frac{555}{220 \times 5} \pm 4\%$$

8.)

$$\sigma_{\text{res}} = \sqrt{\left(\frac{\partial R^2}{\partial R_1}\right)^2 \sigma_1^2 + \left(\frac{\partial R}{\partial R_2}\right)^2 \sigma_2^2}$$

$$= \left(\frac{\partial R^2}{\partial R_1}\right) \Delta R_1^2 + \left(\frac{\partial R}{\partial R_2}\right)^2 \Delta R_2^2$$

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Measurement of energy and Power

5.1) Moving coil in a Dynamometer wattmeter is normally connected "across the supply".

5.2) 230V / 50 Hz AC

$$\begin{aligned} P_{inst} &= V_m \sin \omega t = \frac{V_m \sin \omega t}{R} \\ &= \frac{V_m^2 \sin^2 \omega t}{R} \\ &= \frac{V_m^2}{R} (1 - \cos 2\omega t) \end{aligned}$$

$$2\omega = 2f = 2 \times 50 = 100 \text{ Hz}$$

5.3) $T_D \propto VI \cos \phi$ (active power)

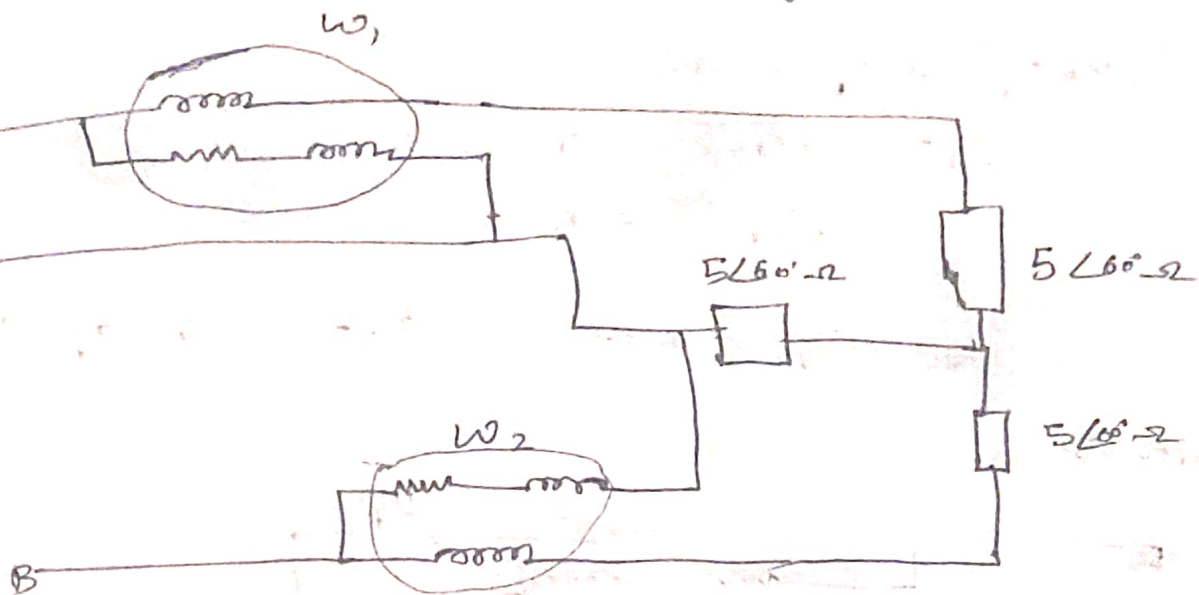
5.4) c) pressure coil are not in phase

5.5) b) Two wattmeter method can take the unbalanced

5.6) Given

$$f = 50 \text{ Hz}, \quad A.C. = V_L \Rightarrow 100 \text{ V}_{rms}$$

R x B sequence.



Wattmeter method read = ?

$$V_L = V = 100 \text{ V} \quad V_{ph} = \frac{100}{\sqrt{3}}$$

$$I_L = I_{ph} = \frac{V_{ph}}{Z} = \frac{100}{\sqrt{3} \times 5 \angle 60^\circ} = \frac{20}{\sqrt{3}} \angle -60^\circ \text{ A}$$

$$\begin{aligned} W_1 &= V_L I_L \cos(30^\circ - \phi) \\ &= 100 \times \frac{20}{\sqrt{3}} \cos(30 - (-60)) \\ &= \frac{2000}{\sqrt{3}} \cos 90^\circ \Rightarrow 0 \text{ W} \end{aligned}$$

$$\begin{aligned} W_2 &= V_L I_L \cos(30^\circ + \phi) \\ &= 100 \times \frac{20}{\sqrt{3}} \cos(30 + (-60)) \\ &= 100 \times \frac{20}{\sqrt{3}} \times \frac{1}{2} \\ &= 100 \text{ W} \end{aligned}$$

$$W_1 = 0 \text{ W}$$

$$W_2 = 100 \text{ W}$$

5.7)

$V_R \rightarrow$ Reference

$$V_R = V \angle 0^\circ \quad V_Y = V \angle -120^\circ \quad V_B = V \angle -240^\circ$$

$$V_R I_R \cos \phi_R = 400 \text{ W}$$

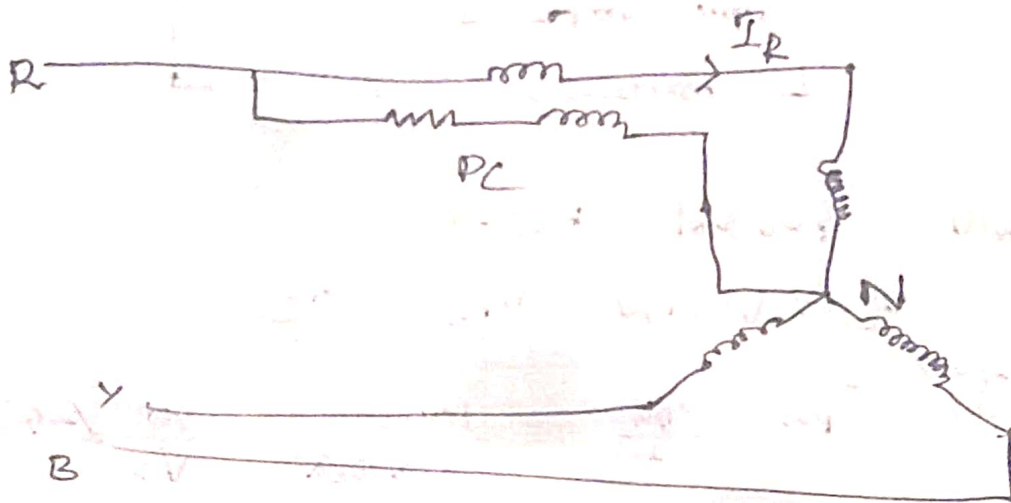
$$V_R I_R \times 0.8 = 400$$

$$V_R I_R = \frac{400}{0.8}$$

$$I_R = I_R \angle -36.87^\circ$$

P.f. = 0.8 lag (inductive load)

Wattmeter



$$V_{RY} = V_{YB} - V_B \rightarrow V \angle -120^\circ$$

$$V \angle -240^\circ$$

$$\Rightarrow \sqrt{3} V \angle -40^\circ$$

$$I_R = I_R \angle -36.87^\circ \text{ A}$$

Angle b/w V_{RY} & I_R

$$\theta = -40 - (-36.87) = -3.13^\circ$$

$$\Rightarrow \sqrt{3} V I_R \cos(-53.13)$$

$$\sqrt{3} \times 500 \times 0.6 \Rightarrow 519.6 \text{ W}$$

$$M' = \frac{1}{M} = \frac{1}{14.14} = \frac{\text{rev}}{\text{sec}}$$

$$M' = \frac{1}{14.4} \frac{\text{rev}}{\text{sec}} \times \frac{1}{0.50} \times 10^3 = \frac{3600}{1 \text{ hr}} \\ = 1000 \text{ rev/hr}$$

$$5.18) P_1 + P_2 = 10.5 - 2.5 = 8 \text{ kW}$$

$$\cos \phi = \cos \left[\tan^{-1} \sqrt{3} \left(\frac{P_1 - P_2}{P_1 + P_2} \right) \right] \\ = \cos \left[\tan^{-1} \left[\frac{10.5 - (-2.5)}{10.5 + 2.5} \right] \right]$$

$$\cos \phi = 0.334$$

$$5.19) \text{ positive power } (+) = \text{negative power } (-) \\ \text{So avg power} = 0 \text{ W}$$

$$5.18) V_c = E_2 \sin \omega t + E_2 \sin 3\omega t.$$

$$i(t) = I_1 \sin(\omega t - \phi_1) + I_3 \sin(\omega t - \phi_3) \\ + I_3 \sin 3\omega t.$$

$$\text{Average power} = \frac{1}{2} E_1 I_1 \cos \phi_1 + \frac{1}{2} I_3 \cos \phi_3$$

$$5.19) W_2 = V_L I_L \cos(30^\circ + \phi)$$

$$W_2 = V_L I_L \cos(30^\circ - \phi)$$

$$W_2 = V_L I_L \cos 60^\circ = \frac{V_L I_L}{2}$$

$$W_L = V_L I_L \cos 30^\circ = V_L I_L$$

$$W_2 = 2 W_1$$

$$5.20) W_1 = 250 \text{ W}, W_L = 100 \text{ W}$$

$$\phi = \tan^{-1} \left(\sqrt{3} \left(\frac{W_1 - W_2}{W_1 + W_2} \right) \right)$$

$$\phi = \tan^{-1} \left(\sqrt{3} \left(\frac{250 - 100}{250 + 100} \right) \right)$$

$$\phi = 36.58^\circ$$

$$\cos \phi = 0.8023$$

5.21) $\cos \phi = 0.2$ LPF wattmeter.

$$W_m = 150 \text{ W}$$

$$V = 150 \text{ V}, I = 10 \text{ A}$$

$$W = VI \cos \phi = 150 \times 10 \times 0.2 = 300$$

$$m = \frac{W}{W_m} = \frac{300}{150} = 2$$

5.22) $\text{pf} = 0.707, \phi = \cos^{-1}(0.707) = 45^\circ$
 $P = 5 \text{ kW}$

$$W_1 + W_2 = 5 \text{ kW}$$

$$\tan \phi = \sqrt{3} \left(\frac{W_1 - W_2}{W_1 + W_2} \right)$$

$$W_1 + W_2 = \sqrt{2} (W_1 - W_2)$$

$$W_1 - W_2 = \frac{5}{\sqrt{3}}$$

$$W_1 = 3.94 \text{ kW}, W_2 = 1.06 \text{ kW}$$

5.24)

$$\phi = \cos^{-1}(-1/2)$$

$$\Rightarrow 120^\circ$$

Power Diagram

$$\phi = \phi - 90^\circ \Rightarrow 30^\circ$$

" I_R leads V_R "

$$\text{Pf} = \cos \phi = \cos 30^\circ = 0.866 \text{ leading}$$